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10

Research Topic: The Effect of Bio-Inspired Leading-Edge Serrations (Tubercles) on the Lift-to-Drag Ratio and Stall Behavior of Small-Scale UAV Wings at Low Reynolds Numbers

The Effect of Bio-Inspired Leading-Edge Serrations (Tubercles) on the Lift-to-Drag Ratio and Stall Behavior of Small-Scale UAV Wings at Low Reynolds Numbers

Abstract

Many small fixed-wing UAVs fly at low Reynolds numbers, in which case the flow of air separates much more easily from the wing surface than it would on a larger aircraft. This separation results in increased drag, reduced lift, and even sudden stall in low-speed maneuvers, turns, takeoff, and landing phases. In this research, the hypothesis is that leading-edge tubercles, rounded sinusoidal teeth found on the leading edges of humpback whale flipper wings, can be used as a passive technology for increasing wing efficiency on small fixed-wing UAVs. There will be two wings tested against a control wing without tubercles in a low-speed wind tunnel. All three wings will have identical characteristics except for the shape of the leading edge. The dependent variables for the study will include lift coefficient, drag coefficient, lift-to-drag ratio, maximum lift coefficient, stall angle, and post-stall lift retention. Testing will take place at Reynolds numbers around 50,000 and 100,000 at angles of attack from −4° to 24°. It is hoped that the study will confirm that leading-edge tubercles provide sufficient stall tolerance to justify any drag penalties they may cause at lower angles of attack. It is anticipated that the tubercled airfoil will stall in a gentler manner and with higher post-stall lift retention, while the smooth airfoil will have the highest lift-to-drag ratio at low angles of attack.

Index Terms: Bio-inspired design, leading-edge tubercles, lift-to-drag ratio, low Reynolds number, stall behavior, unmanned aerial vehicle (UAV) wing aerodynamics.

I. INTRODUCTION

Small fixed-wing UAVs are used for surveys, agriculture, environmental surveillance, photography, inspection, search and rescue, and educational purposes in engineering. Most of the above applications involve the use of UAVs operating at a relatively slow speed with short wing chords. This creates a low value of Reynolds number, ranging from about 10,000 to 200,000. With a low Reynolds number, the boundary layer is easily separated, leading to increased drag, lower lift, and sudden stall.

Stall represents a critical safety issue and design challenge when using UAVs. In the case of low-speed flying or during sharp maneuvers, the wing reaches the critical angle of attack. On a smooth conventional wing, the flow will be easily separated, creating a sudden loss of lift. A wing whose stall is gradual will increase the controllability of the plane during take-off, landing, low-speed turns, and gust recovery.

One possible solution is based on humpback whale flippers. Humpback whales have rounded bumps called tubercles along their leading edges. Fish and Battle described this unusual flipper geometry and suggested that it supports high-angle-of-attack maneuvering [1]. Later, Miklosovic et al. conducted wind-tunnel testing of an idealized humpback-whale flipper and reported that tubercles delayed stall by approximately 40 percent [2].

In aircraft and UAV research, tubercles are commonly modeled as sinusoidal waves along a wing's leading edge [10]. The waves may create streamwise vortices and localized regions of attached airflow [12]. Instead of the entire wing stalling simultaneously, some parts of the wing may continue generating lift while other parts separate. However, tubercles may also add drag or reduce low-angle aerodynamic efficiency [13]. Therefore, this research will investigate whether the potential stall-control benefit outweighs the efficiency penalty for a small-scale UAV wing.

II. RESEARCH QUESTIONS AND HYPOTHESES

This study aims to investigate the influence of leading-edge tubercles on aerodynamic performance and stall characteristics of a small UAV wing at low Reynolds numbers.

Research Question 1: In what ways do leading-edgetuberclese influence lift coefficient (CL), drag coefficient (CD), and lift-to-drag ratio (CL/CD), in comparison to a smooth wing?

Hypothesis 1: Tubercled wings would exhibit different lift and drag characteristics than the smooth wing. At low angles of attack, the smooth wing would have a better lift-to-drag ratio due to a smooth leading edge, which causes less flow separation and drag.

Alternative Hypothesis 1: At moderate to high angles of attack, one or both tubercled wings would have a greater lift-to-drag ratio than the smooth wing due to the possibility of reduced flow separation and loss of lift due to tubercles. There are studies indicating that leading-edge tubercles affect lift, drag, and flow separation.

Research Question 2: Do the tubercled wings achieve a higher maximum lift coefficient (CL, max) or a larger stall angle than the smooth wing?

Hypothesis 2: The tubercled wings will stall at a higher angle of attack and will also show a slower decline in lift following the achievement of maximum lift than the smooth wing baseline.

Alternative Hypothesis 2: The tubercled wings will not produce a higher maximum lift coefficient than the smooth wing. Instead, the key advantage of the tubercles on the wing may be the slower, more gradual stall and lift retention beyond the point where the smooth wing starts to lose lift.

Research Question 3: At what tubercle amplitude does the optimal balance of lift-to-drag ratio and stall control exist?

Hypothesis 3: The low-amplitude tubercled wing will have the optimal balance since it is likely to generate sufficient flow control to defer or dampen the stall effect without incurring a high drag cost before stall with high-amplitude tubercles.

Alternative Hypothesis 3: The high-amplitude tubercled wing will have the strongest stall effect, including the highest stall angle and maximum stall lift retention. However, it will have the lowest lift-to-drag ratio before stall because the bigger tubercles will generate greater air disturbance.

III. LITERATURE REVIEW

The biological background of this study stems from humpback whale flippers. Fish and Battle identified the leading-edge tubercles of humpback whale flippers and suggested that these may be instrumental in maneuvering in water [1]. The study motivated the examination of tubercles as a passive flow control device on man-made wings.

One of the earliest influential wind tunnel experiments carried out with respect to tubercles is that of Miklosovic et al. [2], who demonstrated that an idealized flipper design with leading-edge tubercles had a stall delay of around 40 percent compared to a smooth design.

This study is concerned with the effect of tubercles at low Reynolds numbers, which are computed as:

$$Re = \rho V c / \mu \quad (1)$$

where ρ is the air density, V is the airspeed, c is the wing chord, and μ is the dynamic viscosity of the air. The wings of small UAVs are vulnerable to low Reynolds number effects such as laminar separation bubbles and premature flow separation.

Tubercles on the leading edge of Selig S1223 and NACA 4415 airfoils were experimentally investigated by Sudhakar et al. at the Reynolds numbers of 100,000, 150,000, and 200,000 [3]. The experiments demonstrated that the tubercles can increase lift, decrease drag, and delay the stall of the Selig S1223 airfoil in certain cases. At the same time, the NACA 4415 showed no comparable improvement. Thus, the tubercles cannot be beneficial automatically; their effect depends on the base airfoil and operating conditions.

Aftab and Ahmad studied, using computational fluid dynamics and wind tunnel testing, the spherical and sinusoidal tubercles of a NACA 4415 airfoil [4]. They stated that the effectiveness of the tubercles greatly depends on their geometrical parameters, and certain configurations can increase lift and L/D ratio at high angles of attack.

Seyhan et al. studied the tubercle designs on SD7032 finite wings at Reynolds number values of 55,000 and 110,000 [5]. The study conducted by Seyhan et al. is useful in small UAV development since it considers a finite wing, unlike other studies which have only considered two-dimensional airfoils. The study findings show that certain tubercles increase lift at low Reynolds numbers.

Gopinathan et al. carried out a numerical study on leading-edge tubercles on a swept wing at low Reynolds number, concluding that the sweep angle of the wing and the airfoil section selected play a role in determining the magnitude of the benefits obtained by the presence of tubercles [6]. In a related wind tunnel experiment, Butt et al. made comparisons between smooth and tubercled airfoils in the form of NACA 0021 and 4412 airfoils and showed significant differences in lift characteristics and post-stall characteristics for the two types [7]. Fan et al. carried out numerical and experimental investigations on bionic airfoils at low Reynolds numbers and showed that the amplitude of tubercles is more important than the wavelength of tubercles when it comes to stall delay [8]. In an extension of their previous study on swept wings, Gopinathan et al. showed that the benefits due to the presence of tubercles diminish with increasing sweep angle of the wing, which supports the use of a simple unswept wing geometry for studying the leading-edge effects of tubercles [9].

Tubercles can help to enhance stall performance, although it does not guarantee an improvement in cruise performance according to the scientific literature [1] – [10]. The suggested study seeks to tackle this design challenge by comparing smooth and tuberculated wings using identical airfoils and sizes.

IV. PROPOSED METHODOLOGY

A quantitative experimental methodology will be employed for this study. Three small finite-wing models will be constructed through the use of computer-aided design software, with the same airfoil, span, chord, planform, materials, and surface finish, with the exception of the leading edge.

The chosen airfoil will be the Selig S1223, which is a high-lift airfoil suitable for low-speed and low Reynolds number UAVs. The wing planform will be rectangular and unswept, such that wing sweep will not affect the leading-edge comparison, as sweep angle may hide the influence of the geometry of the tubercles [9].

TABLE I - Wing Model Leading-Edge Configurations

Model	Leading-Edge Configuration
Wing A	Smooth baseline wing (no tubercles)
Wing B	Low-amplitude tubercles, A/c = 0.025
Wing C	High-amplitude tubercles, A/c = 0.05

Sinusoidal tubercles would be used, with a constant wavelength of $\lambda/c = 0.25$, where A is the amplitude of the tubercle, λ is the wavelength, and c is the chord of the wing. The wings would be of approximately equal chord length (0.10 to 0.15 m) and wingspan (0.45 to 0.60 m), dependent on wind tunnel size. The ranges of tubercle amplitudes and wavelengths for Wings B and C are based on previous experimental studies investigating the effect of several amplitude and wavelength values on NACA airfoils [13] and numerical analysis demonstrating that the amplitude is more significant in stall delay compared with the wavelength [8].

Wings would be placed in a low-speed wind tunnel at an adjustable angle of attack. A force balance or load cell would measure the lift and drag forces. Airspeed would be measured using wind tunnel equipment, a Pitot-static tube, or an anemometer. Experiments would be conducted at Reynolds numbers of about 50,000 and 100,000. Wind tunnel calibration, force measurement protocol, and correction for wall and support interference would be in accordance with standard low-speed wind tunnel experiments [14].

Tests on the wings will be conducted from −4° to 24° angle of attack. Data collection will be done every 2° but near the anticipated stall range, it will be done every 1°. During each test, lift force, drag force, airspeed, angle of attack, and air temperature will be noted. Each test will be done at least three times. Yarn tufts or smoke flow visualization will be employed, if possible, in order to record the formation and development of separation because it was found in previous numerical studies that the tubercles help flow through the cavities between bumps and postpone separation along the span direction [12].

The lift and drag coefficients are calculated as:

$$CL = L / (½ \rho V^2 S) \quad (2)$$

$$CD = D / (½ \rho V^2 S) \quad (3)$$

The lift-to-drag ratio will be calculated as:

$$CL/CD = L / D \quad (4)$$

The stall angle will be defined as the angle of incidence where the lift coefficient reaches its maximum value, after which it consistently drops. Post-stall lift efficiency will be determined through comparison of the lift coefficient at 20° with the maximum lift coefficient. The mean value, standard deviation, percentage difference, and graph will be utilized in analyzing the data. One-way ANOVA will be used to compare the three wing models if there are enough repetitions.

In case a CFD component is incorporated in future work, the choice of the turbulence model should be done with care, since previous work has indicated that the predicted lift and drag for tubercled wings may differ widely based on the turbulence model chosen at Reynolds number close to 120,000 [11].

V. EXPECTED OUTCOME AND LIMITATIONS

The smooth wing is predicted to yield the most favorable lift-to-drag ratio at low angles of attack since its leading edge creates fewer disturbances and less drag. There could be a slight efficiency reduction before the stall for the tubercled wings.

But the tubercled wings are also predicted to exhibit gentler stall characteristics and improved lift retention after stall, which is in agreement with earlier results showing that tubercle height is an important parameter for stall delay [8]. The low-amplitude tubercled wing might have the best compromise between efficiency and stall delay while the high-amplitude tubercled wing might exhibit more stall control but suffer from greater drag.

Possible sources of errors may include wind tunnel wall effects, force transducer error, wing support interactions, surface roughness, fabrication differences, and airflow uniformity [14]. The findings can only be applicable directly to the wing design, airfoil shape, tubercle pattern, Reynolds number, and test conditions selected. Tubercles have been found to perform differently depending on the wing sweep angle [9] and base airfoil shape [3]; therefore, the findings cannot be generalized to swept wing designs or any other airfoil shape that has not been tested.

VI. CONCLUSION

In this proposal, the question being answered is whether tubercles derived from the flipper of a humpback whale have the ability to enhance the aerodynamic performance of wings of small-scale UAVs with regard to their Reynolds number. This research shall use a flat baseline wing and two types of wings with low-amplitude and high-amplitude tubercles without changing the wing's airfoil, chord, span, shape, material, and testing conditions inside the wind tunnel.

Instead of assuming that tubercles have positive effects on the aerodynamics of a wing, their trade-off between pre-stall aerodynamic efficiency and stall resistance will be measured. It will be found that tubercles can prove to be most important for UAVs that require safer low-speed handling, gentle stall, and retention of lift at higher angles of attack.

To build upon this work, the optimum shape of the tubercle could be analyzed using different amplitudes, wavelengths, airfoil designs, and CFD analysis to obtain a more realistic view of flow separation and vortex shedding. The next step in this project could be to apply the best wing design to a small fixed-wing UAV and fly tests to analyze low-speed stability, turning capability, stall recovery, endurance, and control performance.

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